Exploring the Trade-offs Between Audio Fidelity and Computational Efficiency in a Custom Real-Time Browser-Based Synthesizer

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How do different oscillator anti-aliasing techniques and digital filter architectures influence the relationship between audio fidelity and computational efficiency in real-time browser-based synthesis?

I. Literature Review

Digital signal processing (DSP) refers to the manipulation of signals represented in discrete form, such as those stored or processed by computers. In computer music, DSP provides the theoretical and practical foundation for digitally modelling and recreating the behaviour of analog audio systems, including oscillators and filters. However, accurately translating analog oscillators and filters into the digital domain presents several technical challenges. Specifically, the choice of oscillator anti-aliasing techniques and digital filter architectures introduces different considerations that influence the accuracy, stability, and computational efficiency of the resulting audio. These considerations become particularly important in real-time browser-based synthesis, where audio fidelity must be balanced against computational constraints.

This section examines these challenges through a review of existing literature, focusing on the methods or approaches used or proposed to address them.

A. Digital Oscillators

Analog oscillators generate the fundamental voltage signals used in analog synthesisers, while digital oscillators aim to reproduce equivalent waveforms in the discrete-time domain. Common waveforms include the sine, sawtooth, square and triangle waves, each with distinct harmonic characteristics that contribute to its perceived timbre.

A naive digital oscillator generates waveforms directly from mathematical functions or phase-based calculations. A sawtooth, for example, can be produced with a phase accumulator, which advances its phase on every sample by an amount determined by the oscillator frequency and the sample rate. The phase is mapped to an amplitude range and, once it reaches its maximum, wraps back to the minimum, producing the characteristic sawtooth shape. This abrupt jump is known as a *discontinuity*, and it introduces infinitely many high-frequency harmonics. The Nyquist–Shannon sampling theorem states that a signal can only be represented accurately if it contains no components above the Nyquist frequency, which is half the sampling frequency. A sawtooth always contains harmonics above this limit, so they cannot be represented and instead fold back into the audible range as inharmonic distortion known as aliasing.

The problem worsens as the fundamental frequency rises. This applies to any waveform with discontinuities, such as the sawtooth and also the square wave, because a greater proportion of their harmonics then lies above the Nyquist frequency.

B. Anti-Aliasing Techniques

Anti-aliasing refers to techniques that reduce the effect of aliasing. One approach is the Band-Limited Step (BLEP) method. The idea is to replace the abrupt discontinuity in the naive waveform with a band-limited approximation, which reduces the high-frequency content introduced by the discontinuity and so improves the accuracy of the eventual reconstruction.

An ideal step is the integral of an impulse, and an ideal band-limited step is therefore the integral of a band-limited impulse, typically a windowed sinc function[1]. Rather than synthesising the entire waveform from band-limited components, the BLEP method computes only the *residual*—the difference between the band-limited step and the naive (non-band-limited) step—and applies it as a correction to the naive waveform in the immediate vicinity of each discontinuity, reducing the effect of aliasing.

A widely used variant is the polynomial BLEP (PolyBLEP), which approximates the residual with a low-order polynomial instead of a table of sinc-derived values[2]. For a normalised phase $t \in [0, 1)$ and a per-sample phase increment $\Delta t = f_0/f_s$, where f_0 is the oscillator frequency and f_s the sampling frequency, a two-sample PolyBLEP residual is given by

$$r(t) = \begin{cases} 2\dfrac{t}{\Delta t} - \left(\dfrac{t}{\Delta t}\right)^2 - 1, & 0 \le t < \Delta t, \\ \left(\dfrac{t-1}{\Delta t}\right)^2 + 2\dfrac{t-1}{\Delta t} + 1, & 1 - \Delta t < t < 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The corrected sawtooth is then obtained by subtracting this residual from the ideal but naive sawtooth waveform $s(t)$

$$y(t) = s(t) - r(t) = (2t - 1) - r(t), \tag{2}$$

which also smoothes the wrap-around over the samples adjacent to the discontinuity. The naive sawtooth wave is chosen to

be $s(t) = 2t - 1$ for simplicity of residual calculation and thus implementation of PolyBLEP, because it reaches its minimal extreme of $s(t) = -1$ at $t = 0$, and the maximum extreme of $s(t) = 1$ at $t = 1$; the discontinuities exist at integer intervals.

A square wave may be treated as the sum of two sawtooth-like discontinuities, one rising and one falling, so the same residual is applied at both transitions (with opposite sign). A triangle wave, being continuous, has no step discontinuities, but its first derivative is discontinuous; this slope discontinuity can be corrected with an analogous polynomial band-limited ramp (PolyBLAMP) residual[3]

The principal advantages of PolyBLEP and PolyBLAMP are their low computational cost, the lack of lookup tables, and suitability for real-time use with modulated frequency, since the correction depends only on the current phase and phase increment. Its limitation is that a low-order polynomial only approximates the ideal band-limited step, so some residual aliasing remains, particularly at high fundamental frequencies, where it is attenuated rather than eliminated.

Alternative approaches offer different trade-offs. Oversampling reduces aliasing by generating waveforms at higher sampling rates before filtering and decimation, but increases computational cost [4]. Wavetable synthesis uses precomputed band-limited waveforms to suppress harmonics above the Nyquist frequency, but requires additional memory and table selection, reducing flexibility especially during dynamic modulation[5].

C. Digital Filter Design Architectures—Biquad Filter

Filters shape the harmonic content of an oscillator's output. A lowpass filter attenuates components above a cutoff frequency, a highpass filter attenuates those below it, and a bandpass filter passes a band around the cutoff while attenuating components on either side. In a synthesizer, the cutoff frequency and resonance are typically modulated in real time, so a digital filter must remain stable and accurate while its coefficients change from sample to sample.

The biquad is the most widely used digital filter structure. It is a second-order infinite impulse response (IIR) filter with a transfer function

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{3}$$

whose coefficients are computed from the cutoff frequency, resonance and filter type, commonly using the closed-form expressions of the Audio EQ Cookbook[1]. Its advantages include low computational cost and the flexibility to obtain different filter responses through changes to the numerator coefficients.

However, its conventional difference-equation implementation presents some challenges under rapid real-time parameter modulation. Changes to the cutoff frequency or resonance require recalculation of the filter coefficients, while the internal delay states retain values generated under the previous coefficient configuration. This mismatch can introduce transient artefacts during modulation[6].

[1]https://www.w3.org/TR/audio-eq-cookbook/

D. Digital Filter Design Architectures—State Variable Filter

To address some of these limitations, the State Variable Filter (SVF) can be implemented using a zero-delay feedback (ZDF) topology and a topology-preserving transform (TPT). Rather than implementing the filter as a conventional difference equation by remembering past outputs, this structure instead remembers integrator-equivalent states to calculate the lowpass, bandpass and highpass responses. Specifically, the trapezoidal integration approximation method allows the feedback structure to be discretised without introducing an additional unit-sample delay into the feedback path[7]. Taken from the same source[7], the analog SVF has the transfer functions

$$H_{\mathrm{LP}}(s) = \frac{1}{s^2 + 2Rs + 1} \tag{4}$$

$$H_{\mathrm{BP}}(s) = \frac{s}{s^2 + 2Rs + 1} \tag{5}$$

$$H_{\mathrm{HP}}(s) = \frac{s^2}{s^2 + 2Rs + 1} \tag{6}$$

where s is normalised to the cutoff frequency and $R = 1/(2Q)$ is the damping factor. Applying the trapezoidal rule to each integrator, with the cutoff frequency pre-warped to compensate for the nonlinear frequency mapping introduced by the discretisation, gives

$$g := \tan\left(\frac{\pi f_c}{f_s}\right) \tag{7}$$

so that the feedback loop can be solved for the current input sample $x[n]$ without a unit delay applied onto the outputs. Specificially, the highpass output is obtained in closed form

$$y_{\mathrm{HP}}[n] = \frac{x[n] - (2R + g)\, s_1[n-1] - s_2[n-1]}{1 + 2Rg + g^2} \tag{8}$$

where notice that each new output does not rely on the past outputs of the filter, and s_1 and s_2 denote the states of the two integrators. The bandpass and lowpass outputs follow from the two integrators in sequence

$$y_{\mathrm{BP}}[n] = g\, y_{\mathrm{HP}}[n] + s_1[n-1] \tag{9}$$

$$y_{\mathrm{LP}}[n] = g\, y_{\mathrm{BP}}[n] + s_2[n-1] \tag{10}$$

Finally, the integrator states are updated for the next sample by the following closed forms

$$s_1[n] = y_{\mathrm{BP}}[n] + g\, y_{\mathrm{HP}}[n] \tag{11}$$

$$s_2[n] = y_{\mathrm{LP}}[n] + g\, y_{\mathrm{BP}}[n] \tag{12}$$

so that $y[n+1]$ will be calculated from these new state values. To be clear then, the ZDF SVF remains an IIR filter—in the sense that state from previous samples are preserved for the calculation of new samples—but its topology offers advantages over conventional biquad difference equation implementations when filter parameters are modulated rapidly, improving filter stability.

II. Methods

This section describes the full approach used to answer the research question: how audio fidelity trades off against computational efficiency in a real-time, browser-based synthesizer. The approach has two parts. First, a synthesizer, *picoDAW*[2], was implemented so that every algorithm under study (naive, PolyBLEP and PolyBLAMP oscillators; Biquad IIR and ZDF state variable filters) runs inside the same signal path, on the same platform, under the same real-time constraints. Second, three controlled experiments were run on that synthesizer, measuring CPU cost, aliasing and filter stability. A final, cross-cutting question is how well these algorithms hold up within the Web Audio API and AudioWorklet environment.

Building a single shared platform, rather than benchmarking isolated reference code, is deliberate. Computational cost in a browser depends on the WebAssembly (WASM) compilation, the AudioWorklet scheduling model and the 128-sample block size, so a measurement taken outside this environment would not answer the question as posed.

A. Implementation of a Browser-Based Synthesizer

1) Platform and development process: picoDAW was implemented with the iPlug2[3] framework, compiled to WebAssembly with Emscripten. The project was built on top of the example instrument provided with iPlug2 (`IPlugInstrument`). The original goal was a multi-voice, multi-instrument synthesizer, but this proved too difficult within the project timeframe, and the scope was reduced to a single configurable voice architecture. No open-source reference implementation was used during development; a few comparable projects were found only while writing this report. One is `web-synth`[4] by Casey Primozic, notable for this report because they wrote a blog post[5] covering how to debug and profile an AudioWorklet browser-based synthesizer, specifically by using Chrome's built-in tracing tool, which will be discussed later in the Methods section.

A significant part of the effort in completing this project went into understanding how iPlug2's WASM build is structured, in particular how the user interface and the audio code communicate. Components were built in the following order: a MIDI sequencer, a gate sequencer (controlling legato versus retrigger behaviour), and then the DSP components compared in the experiments.

2) Two-thread architecture and parameter pipeline: The WASM build of picoDAW runs on two threads.

- **The main JavaScript thread** runs the UI. It executes the compiled `main()` of `IPlugWasmUI` (emitted by `emcc` as `picoDAW-ui.js`) and handles all graphics and user input.
- **The AudioWorklet thread** is spawned by the main thread before the UI loop starts. It repeatedly invokes the `process` function of the `AudioWorkletProcessor` defined in `picoDAW-processor`. Each invocation calls `processBlock`, which is bound to the C++ `IPlugWasmDSP::processBlock` in the DSP module `picoDAW-dsp.js`.

The two threads share no memory and communicate by message passing through the AudioWorklet's `MessagePort`. For example, when the user changes a parameter through the UI, like the gain knob of the synthesizer, `IPlugWasmUI::SendParameterValueFromUI` calls the JavaScript function `setParam(paramIdx, value)` of `IPlugWasmController`, which posts a message of type `'param'` to the worklet:

```
setParam(paramIdx, value) {
  this.workletNode?.port.postMessage({
    type: 'param', paramIdx, value });
}
```

On the other side—the DSP AudioWorklet thread—a `_onMessage` handler checks the message type received on its port, and for `param`, calls `onParam`. Emscripten binds this to `IPlugWasmDSP::OnParamMessage`, which calls `IPlugAPIBase::SetParameterValue` and then the user-level `picoDAW::OnParamChange`. Finally, the message is routed to the actual DSP object through `mDSP.SetParam`, changing the parameters of the underlying filters and oscillators in response, for example. Understanding this pipeline was a prerequisite for the later experiments.

3) DSP structure and voice design: The DSP code is organised in three main files.

a) `picoDAW_DSP.h`: treats the DSP as a black box. It owns a `MidiSynth` instance (`mSynth`) that can hold multiple voices, the smoothers for parameters that are not set per voice, and a low-frequency oscillator (LFO).

b) `picoDAW_Voice.h`: defines a voice, which holds one instance of every algorithm under comparison and selects among them at run time through index parameters:

- **Oscillator cores** (via the `IOscillator` interface): the iPlug2 sine core; custom naive sawtooth, triangle and square cores; and custom band-limited sawtooth, triangle and square cores (PolyBLEP for the discontinuous waveforms, PolyBLAMP for the triangle).
- **Filter cores** (via the `IFilter` interface): the ZDF state variable filter, reworked from the `SVF.h` filter included in IPlug2, and a custom Biquad IIR filter, both with three states set by `mFilterTypeIdx`; lowpass, highpass or bandpass. An IIR Moog ladder filter and ZDF ladder filter were also implemented, but are not discussed in this report.

A voice is thus configured by setting `mSynthCoreIdx`, `mAntiAliasing`, `mFilterCoreIdx`, `mIsZDF`, and the filter cutoff and resonance. Additionally, two ADSR envelopes were added to the signal chain for musical expression; one on the voice amplitude, and one on the filter cutoff. The custom oscillators and filters can be found implemented in `picoDAW_Oscillator.h` and `picoDAW_Filter.h` respectively.

All of these are set through the parameter pipeline above. Because every algorithm sits behind a common interface,

[2]picoDAW
[3]iplug2
[4]https://github.com/ameobea/web-synth
[5]https://cprimozic.net/blog/webaudio-audioworklet-optimization/

the experiments can swap one algorithm for another without changing anything else in the signal path. This is what makes the comparisons controlled.

4) Per-sample signal path and modulation: In each sample, the voice computes the oscillator output at the pitch-derived frequency, multiplies it by the amplitude envelope and gain, and passes it through the selected filter. The filter cutoff is modulated by two sources: the filter ADSR envelope and the LFO output. Both modulate in the octave domain:

$$f_c(n) = f_{c,0} \cdot 2^{\,e(n) + d\,\ell(n)} \tag{13}$$

where $f_{c,0}$ is the base cutoff, $e(n)$ the filter envelope value, $\ell(n)$ the LFO value, and d the LFO depth (fixed at 2 octaves). Cutoff and resonance are updated every sample. This is the most demanding case for a filter, because its coefficients change at audio rate, and it is what the filter stability experiment later stresses.

B. Experiment 1: Computational Effiency

a) Aim: Compare the CPU cost of the naive, PolyBLEP and PolyBLAMP oscillators, and of the Biquad IIR and ZDF SVF filters, within the real-time audio path.

b) Method: The execution time of each 128-sample block was measured with Chrome's built-in tracing[6], recorded and inspected in the Perfetto trace viewer, rather than by timing code inside the AudioWorkletProcessor. Instrumenting the worklet manually was avoided because it adds code to the very path being measured and depends on browser timer resolution.

For each configuration, the synthesizer was set to play a single constant note (for example, a high-pitched sawtooth for the anti-aliasing comparison), and a trace of about 30 seconds was recorded in Chrome while the note sounded. In Perfetto, the whole recording was selected and the slices table was filtered to the `AudioWorkletProcessor::Process` slices, one of which is emitted for every 128-sample block. Perfetto reports, for each slice name, the number of calls, the total wall duration and the mean wall duration, and the mean wall duration per call was recorded as the cost of that configuration. At a sample rate of 48 kHz, a 30-second trace contains 11,250 blocks per configuration (the call count in the trace matches this figure), so means are taken over a large number of blocks.

Each `Process` slice contains child slices. Of these, "author script execution" covers the JavaScript and WebAssembly code of the worklet[7] (including the C++ `processBlock`) and is the part affected by the algorithm under test, whereas the remainder is browser overhead. To provide a more realistic view of the overall effect of different algorithms, the complete `Process` duration was considered the key metric.

c) Justification: Cost per block, compared against the block's real-time budget (128 samples $\approx$ 2.67 ms at 48 kHz), shows directly how much headroom an algorithm leaves and so whether it is practical in a browser. The expectation, from the algorithm structures, is that the Biquad IIR is cheapest, since it needs only a handful of multiplications per sample, and the ZDF filter costs more, since it solves an implicit feedback equation per sample. Furthermore, that using PolyBLEP or PolyBLAMP anti-aliasing costs more than not using anti-aliasing at all. The experiment tests whether that expectation holds in WASM and by how much.

[6]https://web.dev/articles/profiling-web-audio-apps-in-chrome

[7]https://cprimozic.net/blog/webaudio-audioworklet-optimization/

C. Experiment 2: Audio Fidelity (Aliasing)

a) Aim: Quantify how much aliasing each oscillator algorithm produces and how well PolyBLEP and PolyBLAMP suppress it.

b) Method: A high note—A6, fundamental frequency of $f_0 = 1760$Hz—was played on the anti-aliased and naive sawtooth core, with the filter bypassed so that the spectrum reflects the oscillator alone. High fundamentals are used because as discussed in the literature review, few harmonics will fit below the Nyquist frequency, so the rest fold back audibly into the spectrum if not anti-aliased. The output was captured and analysed with an FFT, using a Hann window and averaging over several frames.

Specifically, the output samples of the audio processor were recorded directly in the AudioWorklet, before they reach the browser's audio output, and offered to the user as a 30-second mono WAV download; recording begins when the audio context is initialised. Because the capture is lossless and taken before any system-level processing, the spectra reflect the oscillator alone. This is why a download link will appear in picoDAW after 30 seconds of using it (if it hasn't been updated since the time of writing).

In addition to visual inspection, an alias measure—which we define as the energy of the aliasing present—is computed by using the FFT output. Specifcally, any values that fell within three FFT bin widths of an in-band harmonic were discarded, since they cannot be told apart from the wanted signal. The aliasing energy of a recording was then defined as the sum of the squared spectral magnitude in a window of ± 2 bins around each remaining predicted alias frequency:

$$E_{\text{alias}} = \sum_{j} \sum_{b=-2}^{2} P^2[i_j + b] \tag{14}$$

where i_j is the bin nearest the j-th predicted alias frequency and P is the normalised magnitude spectrum. The two cores are compared by the improvement $10 \log_{10}(E_{\text{alias,naive}}/E_{\text{alias,PolyBLEP}})$ in dB.

c) Justification: The FFT directly shows the unwanted foldback components, and the same note played through different cores isolates the effect of the algorithm. The expectation is that the naive waveforms show a high alias floor, and that PolyBLEP (sawtooth, square) and PolyBLAMP (triangle) reduce these lines substantially. Combined with Experiment 1, this gives the cost and the benefit of each method, so the trade-off in the title can be stated quantitatively.

D. Experiment 3: Filter Stability

a) Aim: Compare how the Biquad IIR and ZDF SVF filters behave under extreme modulation, where coefficient updates are most likely to cause instability.

b) Intended method: A controlled version of this experiment would sweep cutoff and resonance rapidly through automation and monitor the output for non-finite samples, unbounded growth, and discontinuities (clicks), recording these automatically in the processor, for each filter in lowpass, highpass and bandpass mode.

c) Method as carried out: Implementing the automated version was not completed within the project timeframe, so a simpler informal stress test was used. The LFO modulating the filter cutoff was raised to audio rate (up to 1000Hz), resonance was set to its maximum (100%), and the base cutoff was lowered, which is the harshest combination of modulation depth, rate and feedback available in the synthesizer. The output was recorded with OBS as part of a screen capture, and the compressed MP3 audio track was extracted and inspected in MATLAB.

d) Limitations: This test is qualitative. Not only is the recording passed through the system audio path and lossy MP3 compression, the Web Audio output is limited to the range $[-1, 1]$, so the recording shows only that the output saturated, and cannot show how far the internal filter state had grown. The capture also passed through the system audio path and lossy MP3 compression. The result therefore establishes that a filter became unstable under extreme settings, but does not measure by how much one filter is more stable than another.

E. New Tools and Techniques Learned

To carry out this work, the following had to be learned from scratch:

- the iPlug2 framework, including its WASM build, the UI/DSP split, and the `IParam` parameter pipeline;
- the Web Audio `AudioWorklet` model and its `MessagePort`-based communication;
- the Emscripten toolchain, including how C++ methods are bound to JavaScript;
- implementation of PolyBLEP/PolyBLAMP band-limiting and of zero-delay-feedback filter structures[2][3][7];
- in-browser performance measurement and spectral analysis techniques.

III. Results

This section reports the three experiments described in the Methods section: computational cost, aliasing, and filter stability. Each is shown with one figure. All measurements use a sample rate of 48 kHz and a block size of 128 samples, so one block must be processed in at most 2.67 ms (2666.67 μs) to keep up with real time.

A. Computational Overhead: Oscillators

Table I compares the mean processing time per 128-sample block for the naive and band-limited (PolyBLEP for the sawtooth and square, PolyBLAMP for the triangle) oscillator cores. The pitch used was a B6—the highest pitch possible in picoDAW—the filter was bypassed, and the trace length is about 30 s, i.e., 11,250 blocks.

The complete `AudioWorkletProcessor::Process` duration, which includes both the DSP calculations and the relevant browser overhead, is used as the key metric, to provide a more realistic view of the overall effect of different algorithms on computational efficiency.

TABLE I: Mean processing time per block (μs) for naive and band-limited oscillators.

Waveform	Naive	Band-limited	Difference	Relative
Sawtooth (PolyBLEP)	54.59	53.54	-1.05	-1.9%
Square (PolyBLEP)	56.38	56.98	$+0.60$	$+1.1\%$
Triangle (PolyBLAMP)	57.01	58.95	$+1.94$	$+3.4\%$

The band-limited oscillators cost almost the same as the naive ones. The largest difference is for the triangle, where PolyBLAMP adds 1.94 μs per block (3.4%). For the square wave the increase is 0.60 μs (1.1%), and for the sawtooth the band-limited core was measured as slightly *faster* than the naive one (by 1.05 μs). All six configurations lie between 53.5 and 59.0 μs, so the highest cost (the PolyBLAMP triangle) uses 2.21% of the 2666.67 μs block budget.

The sawtooth result should not be taken to mean that PolyBLEP is genuinely cheaper than a naive sawtooth. An algorithm that adds correction terms cannot do less work than the one it extends, and the difference of about 1 μs is comparable to the run-to-run variation seen between measurements elsewhere in this section; like in the following comparison between filters. Each value is a single trace, so the safe reading is that the band-limited cores cost about the same as the naive ones, within a few microseconds, and that the triangle is the most expensive of the three.

B. Computational Overhead: Filters

Figure 1 and Table II show the mean processing time per 128-sample block for the Biquad IIR and ZDF SVF filters as the LFO modulating the cutoff is swept from 0 to 50 Hz.

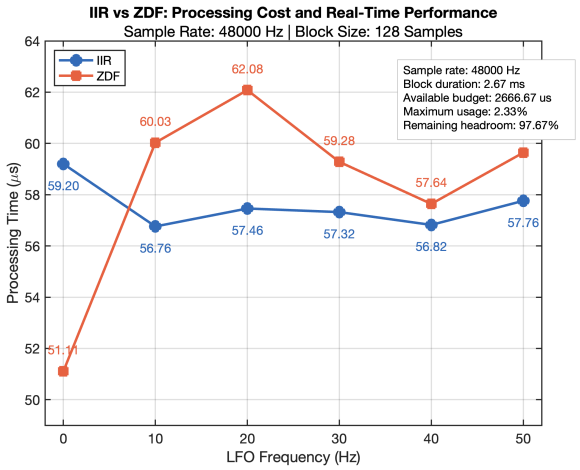


Fig. 1: Mean processing time per 128-sample block for the Biquad IIR and ZDF SVF filters versus LFO frequency (48 kHz sample rate). The maximum recorded cost corresponds to 2.33% of the block budget.

Both filters are very cheap relative to the real-time budget. Every measurement lies between roughly 51 and 62 μs, and the highest value (ZDF at 20 Hz, 62.08 μs) uses 2.33% of the block budget, leaving 97.67% headroom.

TABLE II: Processing time per block (μs) for each filter, and the delta difference

LFO (Hz)	IIR	ZDF	ZDF − IIR
0	59.20	51.11	−8.09
10	56.76	60.03	+3.27
20	57.46	62.08	+4.62
30	57.32	59.28	+1.96
40	56.82	57.64	+0.82
50	57.76	59.64	+1.88

For LFO frequencies from 10 to 50 Hz the ZDF filter is consistently the more expensive of the two, by between about 1 and 5 μs per block (an average of roughly 2.5 μs, or about 4%). The IIR cost is almost flat across LFO frequency (56.8 to 57.8 μs in this range), whereas the ZDF cost varies more, with no monotonic trend.

The 0 Hz point does not follow this pattern: ZDF is cheaper than IIR (51.11 against 59.20 μs). Each point comes from a single trace without repetition, so differences of a few microseconds should be read cautiously, and this point may again reflect run-to-run variation rather than a real effect. This is returned to in the discussion.

C. Aliasing

Figure 2 compares the spectra of the naive and PolyBLEP sawtooth cores at a fundamental of 1760 Hz (top), and the aliasing energy E_{alias} defined in the methods section; see equation (14).

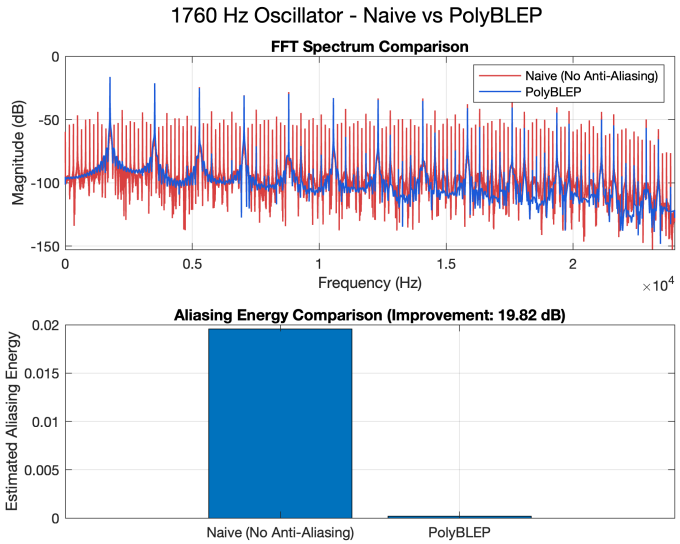


Fig. 2: Top: magnitude spectra of a 1760 Hz sawtooth from the naive and PolyBLEP cores. Bottom: estimated aliasing energy for each core. The improvement is 19.82 dB.

In the spectrum, both cores show the same harmonic peaks at multiples of 1760 Hz. However, the two differ in everything between the harmonics. The naive core shows a dense comb of inharmonic lines across the whole band. These include lines below the fundamental, where an ideal sawtooth has no energy.

These are the harmonics above Nyquist folded back into the audible range; the aliasing. With PolyBLEP, the same lines are mostly absent and the spectrum between harmonics falls to a noise floor of about −90 to −100 dB, an improvement of tens of decibels. Some small residual lines remain around the harmonics, and the upper harmonics of the PolyBLEP signal appear slightly lower than those of the naive signal. The aliasing-energy measure agrees with this. PolyBLEP reduces E_{alias} by 19.82 dB, a factor of about 96 in energy.

D. Filter Stability

Figure 6 shows the output of the ZDF and Biquad IIR filters under the same extreme modulation: an LFO rate of 150 Hz, a base cutoff of 57 Hz, and resonance set to 100%.

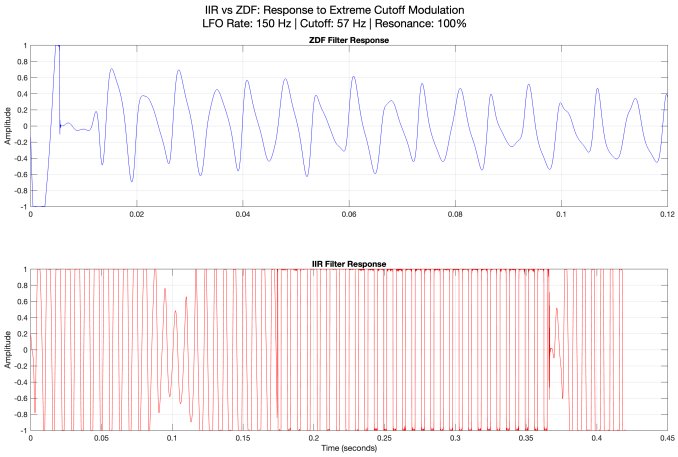


Fig. 3: Filter output under extreme cutoff modulation (LFO 150 Hz, cutoff 57 Hz, resonance 100%). Top: ZDF SVF. Bottom: Biquad IIR.

The two filters behave very differently. The ZDF output shows a brief onset transient, reaching full scale (± 1) for a few milliseconds at the start. It then settles into a bounded, modulated oscillation with peaks of roughly 0.4 to 0.7 and no further saturation over the plotted window.

The IIR output instead repeatedly swings between the limits of -1 and $+1$ and spends most of the plotted time clipped there, giving a square-like waveform. There are short stretches (around 0.09 to 0.12 s and again near 0.37 s) where the amplitude drops below full scale, but it returns to saturation each time. This is the behaviour of a feedback instability, not of a well-behaved filter with a loud signal. In listening, the IIR case produced extremely loud, harsh output, whereas the ZDF case did not.

However, since the output is limited to $[-1, 1]$, the plot shows only that the IIR output saturated, not how far its internal state had grown, a limitation of the experimental method. The effect of ZDF in stabilising the filter output was certainly noticable though; the ZDF switch was even used during this experiment as a panic button to turn off the tremendously loud signal from the IIR filter when pushed to the extreme.

E. Summary of Results

Taken together, the results give the following picture. Anti-aliasing with PolyBLEP reduced aliasing energy by almost

20 dB at 1760 Hz. The ZDF filter cost a few microseconds more per block than the Biquad IIR, but remained stable under modulation that drove the IIR into sustained saturation. All the filter configurations tested used no more than 2.33% of the real-time budget.

IV. Discussion

A. Interpretation of Results

a) Cost versus aliasing: The central finding is that, in this implementation, the fidelity gained from anti-aliasing costs almost nothing in computation. PolyBLEP reduced the aliasing energy of a 1760 Hz sawtooth by a large amount of 19.82 dB (Figure 2), and yet the band-limited oscillators ran within only about 1 to 2 μs per block of the naive ones (Table I). The triangle with PolyBLAMP was the most expensive, at 3.4% above its naive counterpart, but even this used only 2.21% of the real-time block budget. The measured sawtooth result, in which the band-limited core was slightly faster than the naive one, is not evidence that PolyBLEP is cheaper. It is more plausibly measurement variation of the order of 1 μs, and the safe conclusion is that the two have indistinguishable cost at the resolution of these measurements. This agrees with the expectation from the algorithms, since PolyBLEP and PolyBLAMP add a small correction only in the few samples around each discontinuity, so the extra work per block is small[2][3].

b) Filter cost.: The ZDF state variable filter was usually more expensive than the Biquad IIR, by 2.51 μs on average (about 4.4%) across LFO rates of 10 to 50 Hz, which matches the expectation that it would use more cycles. The difference is small, and it is smaller than the variation between repeated measurements would plausibly be. The IIR cost was almost constant across LFO rates, whereas the ZDF cost varied more. The 0 Hz point is an exception, where ZDF was somehow 8.09 μs *cheaper* than IIR.

We did not investigate this. One possible explanation is that the ZDF code does less work when the cutoff is not changing—perhaps values for the `std::tan` are cached, so that `UpdateCoefficients` does zero work on each sample, when the LFO is at 0Hz—but this is a hypothesis, not something the data show. A more mundane possibility is run-to-run variation, since each point is a single trace.

c) Stability.: The stability experiment produced the clearest qualitative contrast (Figure 6). Under an LFO rate of 150 Hz, a base cutoff of 57 Hz and 100% resonance, the Biquad IIR output saturated repeatedly at the limits of the output range, whereas the ZDF output remained bounded after an initial transient. This is consistent with the expectation that a direct-form biquad whose coefficients change at audio rate can inject energy, because its internal state was accumulated under coefficients that have since changed, while a ZDF structure that solves the feedback loop explicitly tolerates such changes better[7]. Because the output is clipped to $[-1, 1]$, the recording shows that the IIR output diverged but cannot show how far the internal state had grown, and it provides no stability margin. The honest summary is therefore that the ZDF filter remained stable *under the conditions tested*, but not that it is unconditionally stable; there could be a situation where both the ZDF and biquad filters are unstable.

d) The trade-off: Combining the three experiments answers the question posed in the title of this report; the measured costs of both anti-aliasing and ZDF filtering are small compared with the real-time budget, and the benefits are large. Specifically, a simple implementation of PolyBLEP and PolyBLAMP produced a reduction in aliasing of almost 20 dB, and the ZDF method produced a filter that stays usable under modulation that makes the cheaper alternative unusable. In this setting there is little tension between fidelity and efficiency. The cheaper algorithms were not meaningfully faster, so choosing them saves almost nothing—in the context of a browser-based synthesizer—while costing obvious audible quality.

B. Feasibility in the Browser

The results also address the browser-specific part of the research question. Every configuration measured used no more than about 2.3% of the 2.67 ms block budget at 48 kHz, so these algorithms are clearly practical in WebAssembly inside an AudioWorklet for a single voice. Two caveats apply. First, the measured times include fixed browser overhead around each call, not just the synthesizer, so the cost of an additional voice would be lower than the full per-block figure. Second, scaling to many voices has not been tested. A simple extrapolation from about 60 μs per block would suggest a large number of voices, but this should not be relied on, since browser scheduling variation, memory behaviour and worst-case spikes (which the mean hides) are likely what causes dropouts in practice[8]. The development experience also showed that the main difficulty in this environment was not DSP cost but architecture: the separation of the UI and audio threads, with parameters passed by message, constrained how modulation and automation could be implemented.

C. Limitations

This report answers parts of the research question more fully than others, and the following limitations should be kept in mind.

- **Single measurements** Each CPU figure comes from one trace, with no repetition. Differences of a few microseconds, including the sawtooth result and the 0 Hz anomaly, are within plausible variation. Reporting block times over repeated runs would make the cost comparisons much firmer.
- **Aliasing coverage** The quantitative alias measure was applied to the sawtooth at a single pitch (1760 Hz). The PolyBLEP square and PolyBLAMP triangle were not assessed with the same measure. The metric is also an absolute energy and not a ratio to the harmonic energy, so it depends on both recordings having the same level.
- **Informal stability test** The stability experiment was a stress test at one extreme setting, not the automated sweep with automatic detection of non-finite values and

[8] https://cprimozic.net/blog/webaudio-audioworklet-optimization/

clicks that was originally planned. It shows qualitative divergence in one filter but does not compare stability across filter modes (lowpass, highpass, bandpass), nor does it quantify margins.

- **Filter response accuracy** The accuracy of the filters' frequency responses was not measured, so that part of the audio fidelity question remains open.
- **Scope** The original aim of a multi-voice, multi-instrument synthesizer was not achieved. All experiments use a single voice, and some parameters (such as LFO rate, sustain and gain) are global rather than per voice.

D. Relation to Previous Work

The results are consistent with the well-established account in the literature that PolyBLEP-style correction suppresses aliasing at low cost[2], and that zero-delay-feedback filters behave better than direct-form structures under fast parameter changes[7]. The contribution of this work is not a new algorithm but a controlled comparison of these methods within a single browser-based system, measured in the environment where they would be used, with a real-time budget that is much tighter and less predictable than in a native plugin.

E. Future Work

The most valuable extensions would be: repeating the CPU measurements and reporting spreads and worst-case times; applying the alias measure to all waveforms and several pitches; replacing the informal stability test with the automated sweep, including detection of non-finite output and clicks, across all filter modes; measuring filter response accuracy against an analogue prototype; and testing polyphony to find the point at which real-time reliability is lost in the browser.

V. Conclusion

This report compared naive, PolyBLEP and PolyBLAMP oscillators, and Biquad IIR and ZDF state variable filters, in a real-time browser-based synthesizer built with iPlug2 and WebAssembly. In this system the fidelity benefits were large and the computational costs small: PolyBLEP reduced aliasing energy by 19.82 dB for a negligible change in processing time, the ZDF filter cost about 4% more than the Biquad IIR, and only the ZDF filter remained stable under extreme modulation. All tested configurations used less than about 2.3% of the available real-time budget for a single voice. These results indicate that, for this application, the more accurate algorithms are the better choice, since the saving from the simpler ones is too small to justify the loss in quality. The conclusions are limited by single measurements, partial coverage of waveforms and filter modes, and an informal stability test, so they should be read as a well-supported first answer to the research question and not as a complete one.

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Appendix

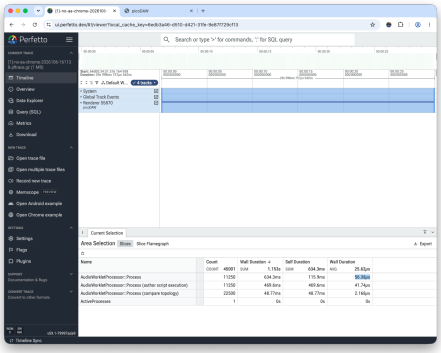


Fig. 4: Measuring computational complexity by using the Chrome browser's built-in tracing, recorded and inspected in the now-default Perfetto trace viewer UI.

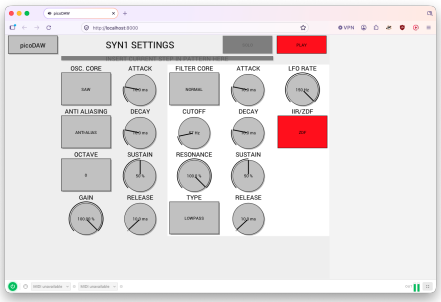


Fig. 5: Settings used to drive the filter output to instability, Experiment 3; LFO 150 Hz, cutoff 57 Hz, and resonance 100%). Notice the switch for switching between IIR and ZDF filter, and the switch also for switching between the naive and anti-aliased oscillator cores.

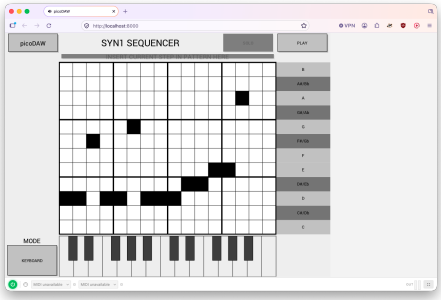


Fig. 6: The MIDI sequencer used for the experiments; by default the bottom control is a keyboard, and can be toggled to show a gate (aka legato/retrigger) sequencer, and a velocity sequencer too.